QQ-A-250/30A
November 19, 1982
SUPERSEDING
QQ-A-250/30
January 9, 1974

FEDERAL SPECIFICATION SHEET

ALUMINUM ALLOY 2219, PLATE AND SHEET

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

The complete requirements for procuring the aluminum alloy 2219 plate and sheet described herein shall consist of this document and the latest issue of QQ-A-250/GEN (see 2.1).

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the specific requirements for aluminum alloy 2219 plate and sheet.

1.2 Classification.

1.2.1 Tempers. The plate and sheet shall be classified as O, T31, T37, T62, T81, T87, T351, T851 or F temper, as specified (see 6.2 and 6.3). The definitions of these tempers shall be as specified in QQ-A-250/GEN.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on date of invitation for bids or solicitation for offers, form a part of this specification to the extent specified herein.

Federal Specifications

QQ-A-250/GEN - Aluminum and Aluminum Alloy Plate and Sheet; General Specification for

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20370.

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(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

3. REQUIREMENTS

3.1 Chemical composition.

3.1.1 The chemical composition shall conform to the requirements specified in table I.

TABLE I. Chemical composition 1/

Element	Percent	
	Minimum	Maximum
Copper	5.8	6.8
Manganese	0.20	0.40
Zirconium	0.10	0.25
Vanadium	0.05	0.15
Titanium	0.02	0.10
Iron	--	0.30
Silicon	--	0.20
Zinc	--	0.10
Magnesium	--	0.02
Others, each	--	0.05
Others, total	--	0.15
Aluminum	Remainder	

1/ Analysis shall regularly be made only for the elements specifically mentioned in table I. If, however, the presence of other elements is indicated or suspected in amounts greater than the specified limits, further analysis shall be made to determine that these elements are not present in excess of specified limits

3.2 Mechanical properties.

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3.2.1 Mechanical properties of material as supplied. The mechanical properties perpendicular to the direction of final rolling, except for material less than 9 inches in width, shall conform to the requirements of table II for the temper specified. For material less than 9 inches in width, the mechanical properties parallel to the direction of final rolling shall conform to the requirements of table II for the temper specified.

TABLE II. Mechanical properties

Temper	Thickness, inches	Tensile strength, minimum, ksi	Yield strength at 0.2 percent offset, minimum, ksi	Elongation in 2 inches [1/] or 4 times D [2/], minimum, percent
O	0.020 - 2.000	32.0 (max)	16.0 (max)	12
T31	0.020 - 0.039	46.0	29.0	8
	0.040 - 0.249	46.0	28.0	10
T351	0.250 - 2.000	46.0	28.0	10
	2.001 - 3.000	44.0	28.0	10
	3.001 - 4.000	42.0	27.0	9
	4.001 - 5.000	40.0	26.0	9
	5.001 - 6.000	39.0	25.0	8
T37	0.020 - 0.039	49.0	38.0	6
	0.040 - 2.500	49.0	37.0	6
	2.501 - 3.000	47.0	36.0	6
	3.001 - 4.000	45.0	35.0	5
	4.001 - 5.000	43.0	34.0	4
T62 [3/]	0.020 - 0.039	54.0	36.0	6
	0.040 - 0.249	54.0	36.0	7
	0.250 - 1.000	54.0	36.0	8
	1.001 - 2.000	54.0	36.0	7
T81	0.020 - 0.039	62.0	46.0	6
	0.040 - 0.249	62.0	46.0	7
T851	0.250 - 1.000	62.0	46.0	8
	1.001 - 2.000	62.0	46.0	7
	2.001 - 3.000	62.0	45.0	6
	3.001 - 4.000	60.0	44.0	5
	4.001 - 5.000	59.0	43.0	5
	5.001 - 6.000	57.0	42.0	4

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TABLE II. Mechanical properties - Continued

Temper	Thickness, inches	Tensile strength, minimum, ksi	Yield strength at 0.2 percent offset, minimum, ksi,	Elongation in 2 inches 1/ or 4 times D 2/, minimum, percent
T87	0.020 - 0.039	64.0	52.0	5
	0.040 - 0.249	64.0	52.0	6
	0.250 - 1.000	64.0	51.0	7
	1.001 - 3.000	64.0	51.0	6
	3.001 - 4.000	62.0	50.0	4
	4.001 - 5.000	61.0	49.0	3
F	All	4/	4/	4/

1/ Not required for material 1/2 inch or less in width
2/ D represents specimen diameter
3/ Material in the T62 temper is a user heat-treated condition and is not available from the material producer
4/ No requirements

3.2.2 Mechanical properties after heat treatment. In addition to conforming to the requirements of 3.2.1, material in the tempers identified in the following paragraphs shall, after having been processed to tempers also identified therein, have properties conforming to those specified in table II, as applicable.

3.2.2.1 Material in the annealed (O) and as-fabricated (F) tempers. Material in the O or F tempers, without the subsequent imposition of cold work or forming operations, shall, after proper solution and artificial aging treatments, develop the properties specified for the T62 temper (see 6.4).

3.2.2.2 Material in the T31, T351, T81 and T851 tempers. Material in the T31, T351, T81 and T851 tempers shall, without the subsequent imposition of cold work or forming operations, be heat treatable to the properties specified for the T62 temper. Such capability shall be demonstrated when specified (see 6.2 and 6.4).

3.2.2.3 Material in the T31, T351 and T37 tempers. Material in the T31, T351 and T37 tempers shall be heat treatable to the properties specified for the T81, T851 and T87 tempers, respectively. Such capability shall be demonstrated when specified (see 6.2).

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3.2.3. Bend test. Bend test specimens taken from material shall be capable of withstanding without cracking the bend test specified in QQ-A-250/GEN. The values for bend factor N are given in table III.

TABLE III. Bend test factor "N".

Thickness, inch	O temper
0.020 - 0.250	4
0.251 - 0.750	6
0.751 - 1.000	8

3.3 Internal defects. When specified (see 6.2), plate shall be ultrasonically inspected (see QQ-A-250/GEN). Acceptance limits shall be as specified in table IV.

TABLE IV. Ultrasonic discontinuity acceptance limits. 1/

Thickness, inches	Maximum weight per piece, pounds 2/	Discontinuity class 3/
0.500 - 1.499	2,000	B
1.500 - 3.000	2,000	A
3.001 - 4.500	2,000	B
4.501 - 6.000	4/	4/

1/ Discontinuities in excess of those listed in table IV may be allowed subject to the approval of the procuring activity, if it is established that they will be removed by machining or that they are in non-critical areas

2/ Class for weights in excess of those shown shall be as agreed upon between the user and supplier

3/ The discontinuity class limits are defined in QQ-A-250/GEN

4/ Shall be as agreed upon by the user and supplier

3.4 Resistance to stress corrosion cracking. Plate 0.750 inch and thicker supplied in the T851 and T87 tempers or in the T351 and T37 tempers and aged to the T851 and T87 tempers shall exhibit no stress-corrosion cracking when subjected to the test specified in 4.3. When specified (see 6.2), plate 0.750 inch and thicker, without prior cold work or forming, when heat-treated to the T62 temper shall exhibit no stress-corrosion cracking when subjected to the test specified in 4.3 (see 6.4).

3.5 Marking. In addition to the marking required in QQ-A-250/GEN, sheet and plate in the T81, T87 and T851 tempers shall be identified by a lot number marked in at least one location on each piece.

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4. QUALITY ASSURANCE PROVISIONS (see QQ-A-250/GEN)

4.1 Heat treatment.

4.1.1 Aging period before testing. Specimens in the T31, T351 and T37 tempers shall not be required to be tested within 4 days after completion of the heat treatment. If the manufacturer so elects, samples may be tested after less than 4 days aging; but if they fail to show the specified properties, the test samples may be discarded and additional specimens may be tested after 4 days aging. These specimens shall be taken from the same locations in the production lot or sample from which the prior specimens were taken.

4.2 Mechanical tests after heat treatment.

4.2.1 Number of tests after heat treatment. From material in each temper of those specified for heat treatment capability demonstrations in 3.2.2 and 6.2, an additional number of specimens equal to that required in QQ-A-250/GEN shall be taken and tested after heat treatment to each temper specified to determine conformance to 3.2.2.

4.3 Stress-corrosion cracking test. Samples of plate in the thicknesses and tempers identified in 3.4 shall be taken in accordance with QQ-A-250/GEN. Test specimens shall be selected in a manner that will permit application of the specified tensile stress in the short transverse direction with respect to grain flow. Specimens shall be stressed in tension in the short transverse direction and be held at constant strain. The stress level shall be 75 percent of the long transverse yield strength. The specimens shall be subjected to the stress-corrosion test specified in QQ-A-250/GEN. It is the responsibility of the contractor that test results be recorded, identified as to lot, and maintained as specified (see 6.2).

5. PREPARATION FOR DELIVERY (see QQ-A-250/GEN)

6. NOTES

6.1 Intended use. Aluminum alloy 2219 plate and sheet is intended for use where elevated temperature strength is required. This alloy is also noted for its good weldability and stress-corrosion resistance in the artifically aged tempers. Use of material in the T31, T351 and T37 tempers is not recommended for finished products when optimum corrosion resistance is required.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number and date of this specification
(b) Form, quantity and temper required (see 1.2.1)

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(c) Dimensions required
(d) Requirements for sizes not specifically covered (see QQ-A-250/GEN)
(e) Whether ultrasonic inspection of plate is required (see 3.3)
(f) Whether stress-corrosion cracking test for plate heat-treated to the T62 temper is required (see 3.4)
(g) Selection of applicable levels of preservation, packaging and packing required, if other than level C (see QQ-A-250/GEN)
(h) Duration of maintenance of stress-corrosion test results
(i) Demonstrations of heat treatability (see 3.2.2.2 and 3.2.2.3)

6.3 Temper. Tempers T351 and T851 are available in plate only.

6.4 Properties after heat treatment. Mechanical and corrosion properties of producer-supplied material are certified for the temper of material supplied. The producer's capability demonstration is not evidence that user-treated material conforms to property requirements of a given temper. Frequently, user-heat-treated material may develop a lower level of properties, especially if any cold, warm, or hot work is introduced prior to solution heat treatment. The user should be held responsible for demonstrating that his processing meets strength and corrosion requirements.

MILITARY INTERESTS:

Custodians

Army - MR
Navy - AS
Air Force - 20

Review Activities

Army - AR
Navy - SH
Air Force - 99
DLA - IS

User Activity

Army - MI

CIVIL AGENCY COORDINATING ACTIVITIES:

NASA - JFK, MSF
GSA - FSS

PREPARING ACTIVITY:

Navy - AS

DOD project 9535-0234

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